

UNIT 3: AI Product Strategy & OpenAI Integration



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- Understanding AI APIs (e.g., OpenAI API – concepts only)
- Basic API key and security awareness
- Cost awareness in AI products (token concept)
- Basic AI evaluation methods

Product lifecycle management



- It is the process of managing an AI product through all its stages, from idea and design to deployment, monitoring, improvement, and retirement.



Stages of AI Product Lifecycle

1. Idea & Problem Identification



- **Purpose**
 - Identify the real-world problem that AI will solve.
- **Activities**
 - Understand user needs.
 - Define project goals.
 - Create problem statement.
 - Identify target users.
- **Example:** Build an AI tutor to help students learn mathematics.

2. Data Collection



- **Purpose**
- Gather quality data to train the AI model.
- **Activities**
- Collect data from surveys, sensors, databases, etc.
- Clean and organize the data.
- Remove duplicate or incorrect records.
- **Example:** Collect student quiz scores and learning history.

3. AI Model Development



- **Purpose**
- Train the AI model using collected data.
- **Activities**
- Select AI algorithm.
- Train the model.
- Improve prediction accuracy.
- **Example:** Train an AI model to recommend personalized study materials.

4. Testing & Validation



- **Purpose**
- Check whether the AI works correctly before release.
- **Activities**
- Test accuracy.
- Detect bias.
- Check security and fairness.
- Validate results.
- **Example:** Test whether the AI tutor gives correct answers.

5. Deployment



- **Purpose**
 - Release the AI product for real users.
- **Activities**
 - Integrate AI into website or mobile app.
 - Make it available to users.
- **Example:** Launch the AI tutor app on the Play Store.

6. Monitoring & Maintenance



- **Purpose**
 - Monitor the AI system after deployment.
- **Activities**
 - Track performance.
 - Fix bugs.
 - Monitor user feedback.
 - Check for model drift.
- **Example:** Fix incorrect recommendations reported by users.

7. Continuous Improvement



- **Purpose**
- Improve the AI system using new data and feedback.
- **Activities**
- Retrain the AI model.
- Add new features.
- Improve accuracy.
- **Example:** AI tutor learns from students' mistakes and provides better explanations.

8. Retirement / Replacement



- **Purpose**
- Remove or replace the AI product when it becomes outdated or no longer useful.
- **Activities**
- Stop using the old model.
- Replace it with a new version.
- Archive important data if required.
- **Example:** Replace an old AI chatbot with a more advanced AI assistant.

Minimum Viable Product (MVP)



- **Minimum Viable Product (MVP)** is the **first basic version** of an AI product that contains **only the essential features** needed to solve the main user problem. It is released to a small group of users to collect feedback before developing the full product.

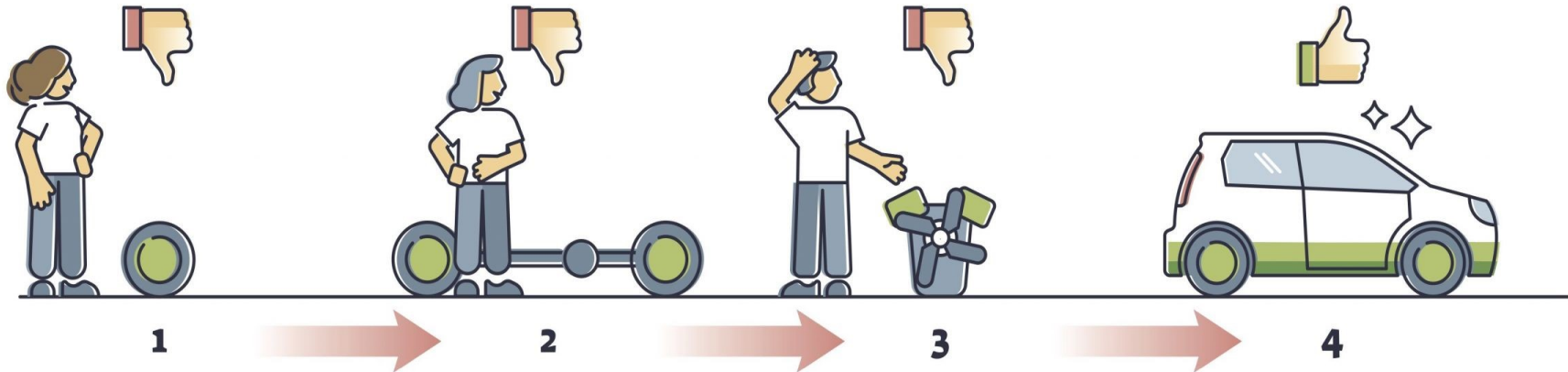
Why is MVP Important in AI Product Design?



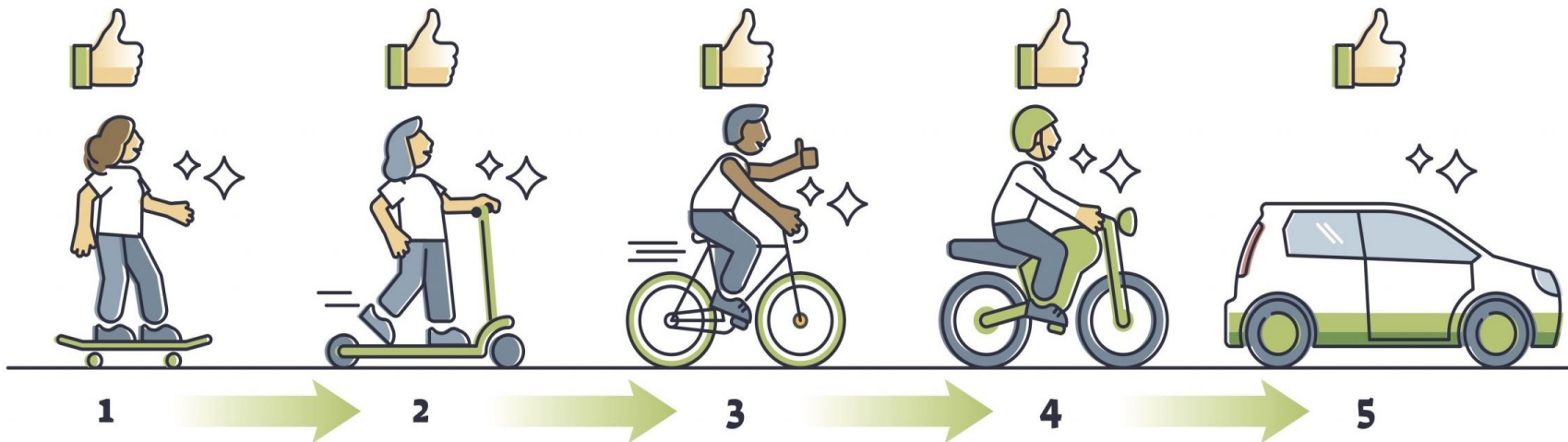
- Validates the AI idea before full development.
- Saves time and development cost.
- Collects real user feedback.
- Identifies problems early.
- Improves the AI model based on actual usage.
- Reduces the risk of product failure.

HOW TO BUILD A MINIMUM VIABLE PRODUCT

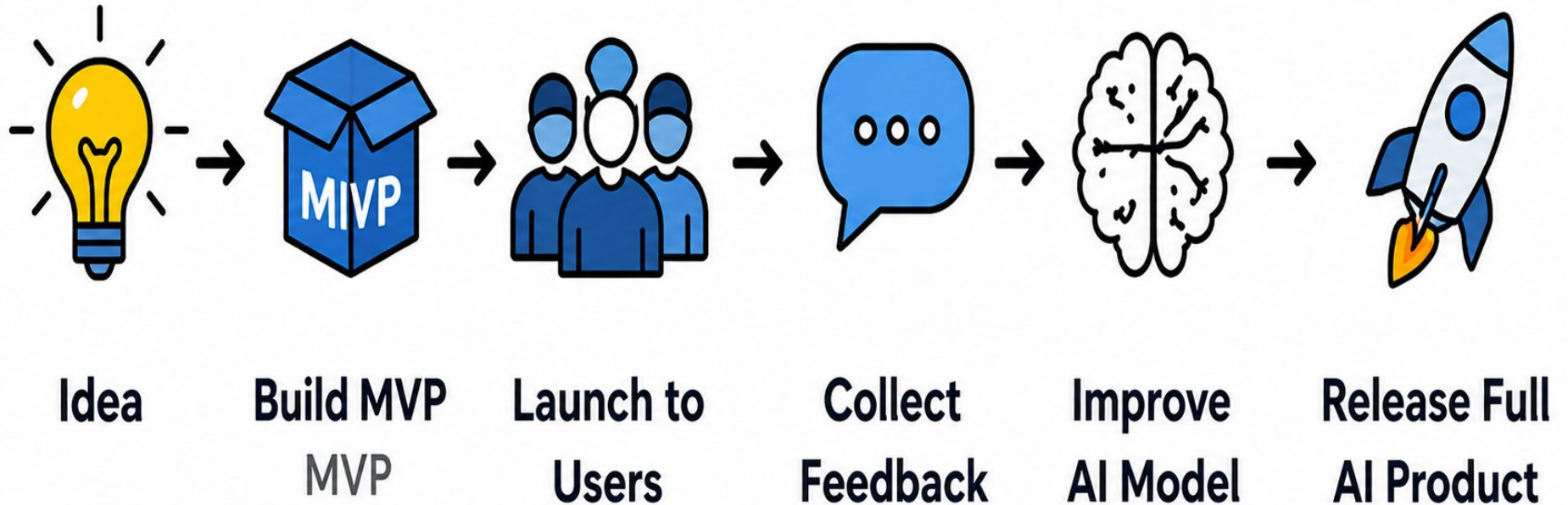
NOT LIKE THIS



LIKE THIS



AI Product Development with MVP



Real-Time Example 1: AI Medical Assistant



- **Problem**

- Patients wait a long time to get basic health information.

- **MVP Features**

- Symptom checker
- Basic health advice
- Hospital locator

- **Future Features**

- Video consultation
- Health report analysis
- Medicine reminders
- Wearable device integration

Real-Time Example 2: AI Travel Planner



- **Problem**
- Planning trips takes too much time.
- **MVP Features**
- Destination suggestions
- Budget estimation
- Simple itinerary generation
- **Future Features**
- Flight booking
- Hotel booking
- Weather alerts
- Voice assistant

Real-Time Example 3: AI Study Assistant

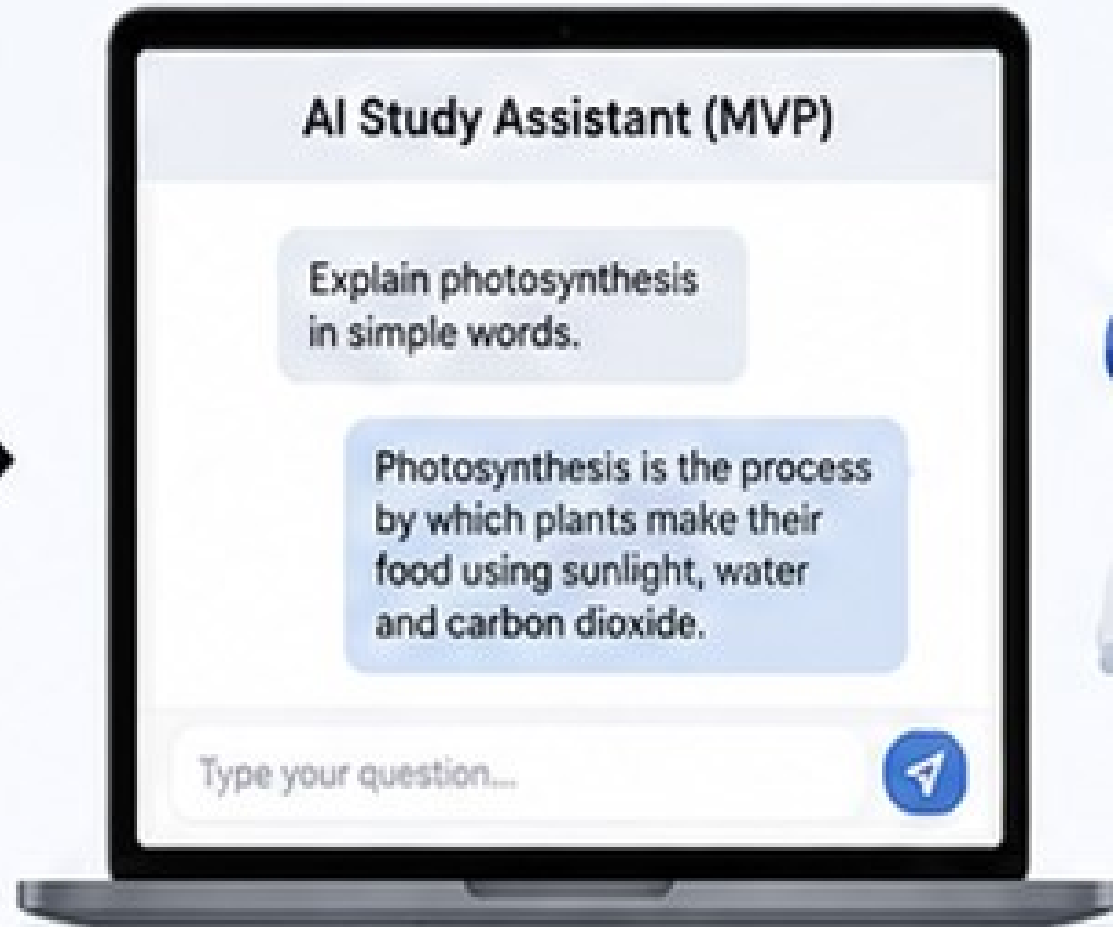


- **Problem**
- Student needs help in studying and organizing their time.
- **MVP Features**
- Answer student questions
- Daily study reminders
- Quiz generation
- **Future Features**
- Voice tutor
- Performance analytics
- Personalized learning plans
- Parent progress reports

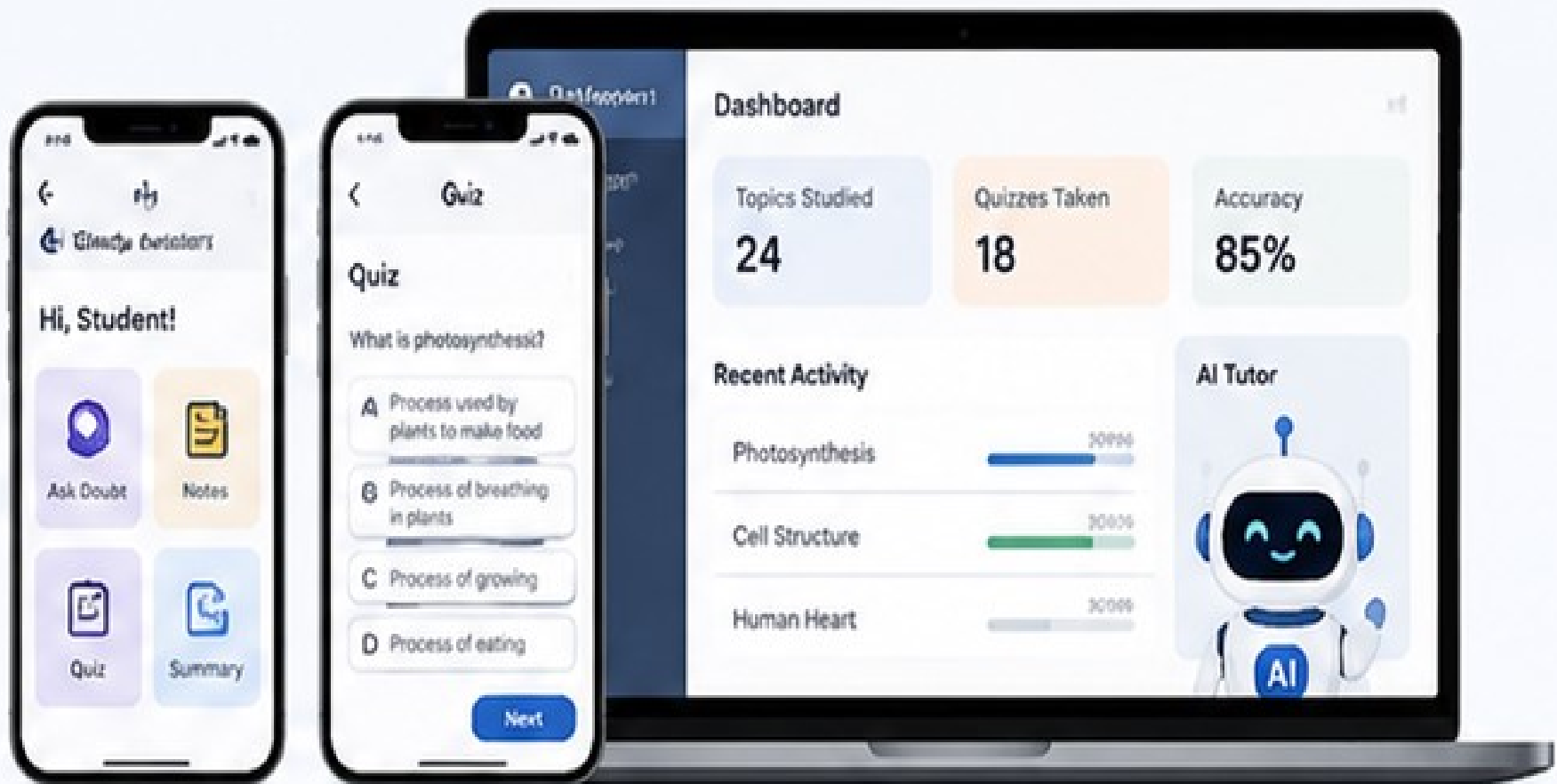
1 Example 1: AI Study Assistant



Problem



MVP

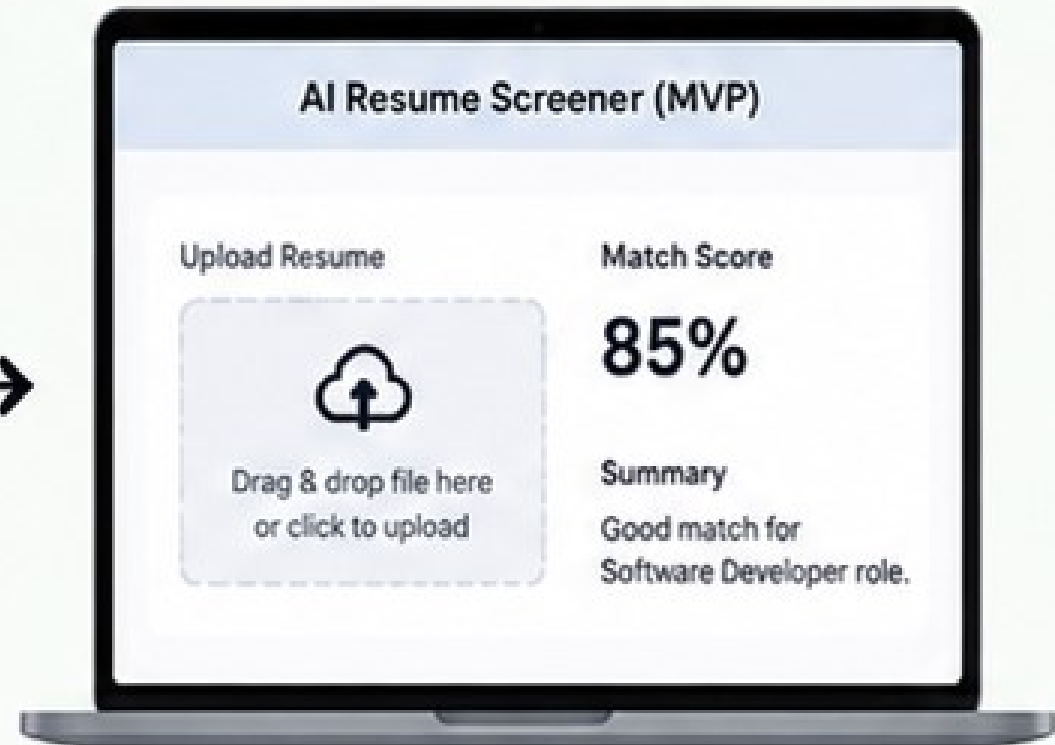


Later Full Product

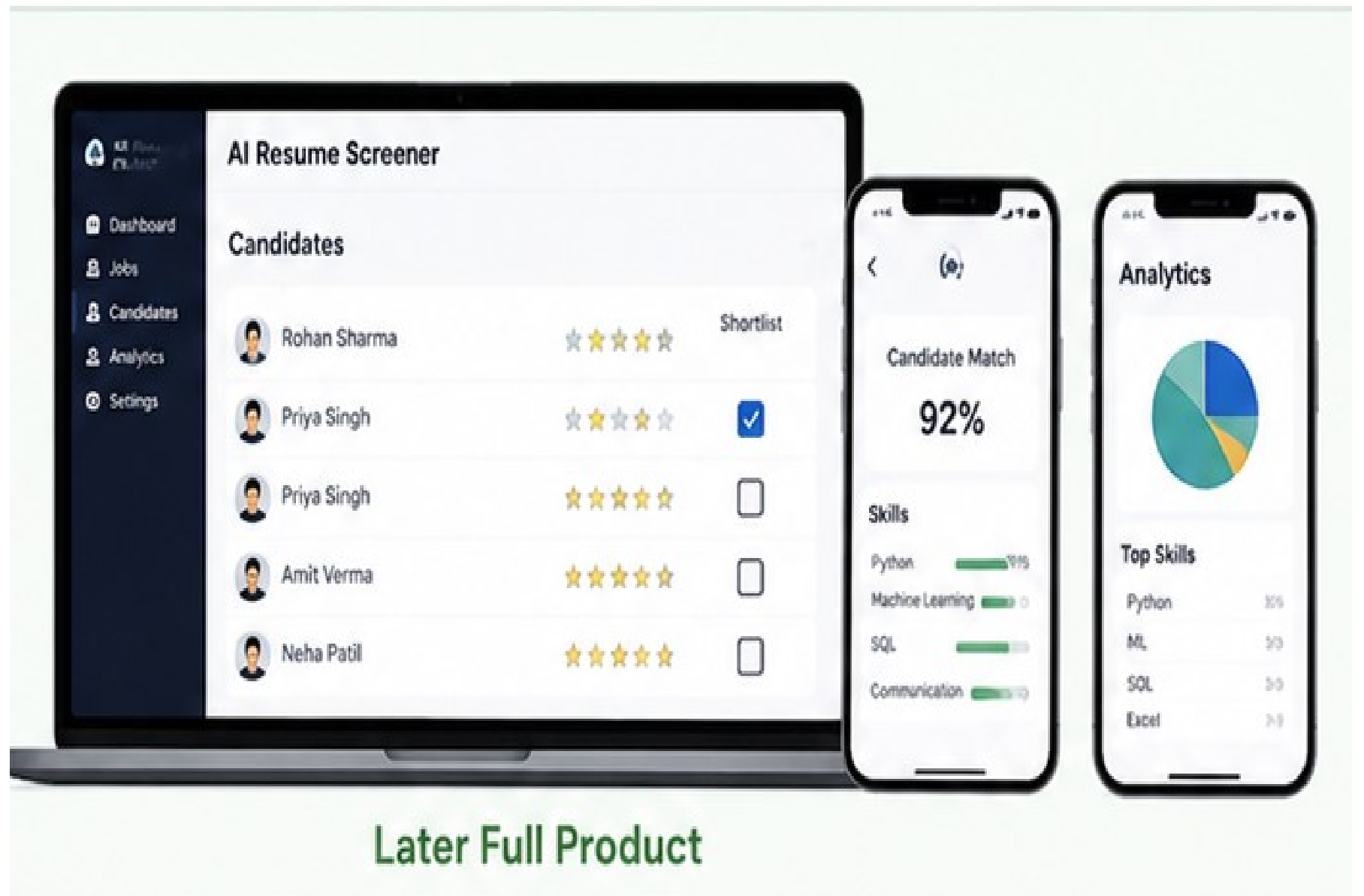
2 Example 2: AI Resume Screening Tool



Problem



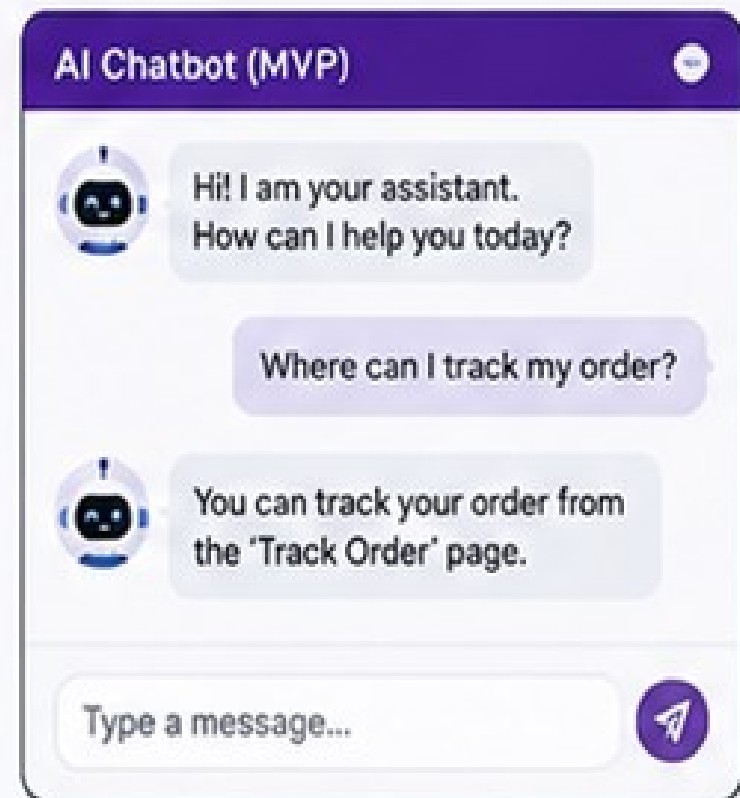
MVP



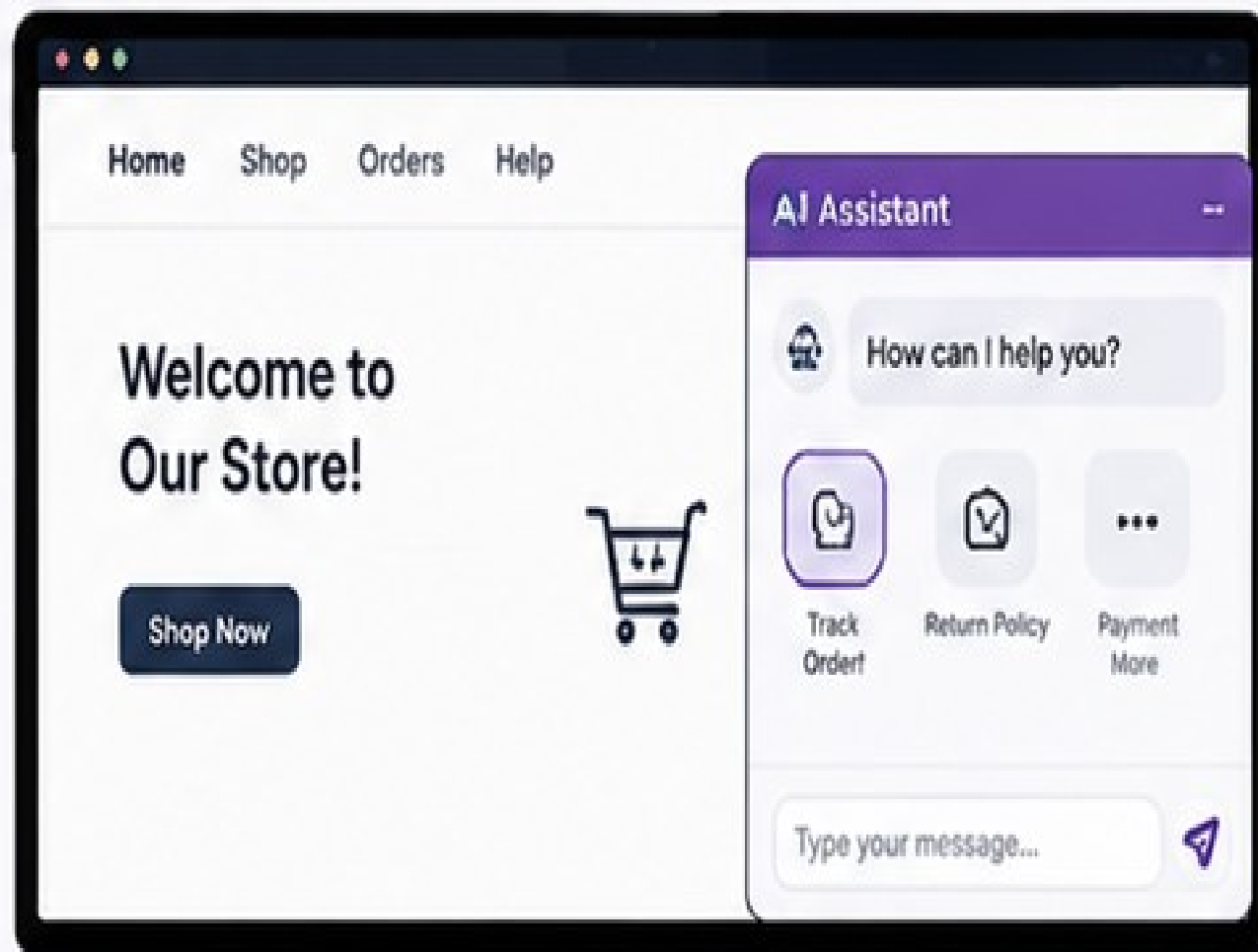
3 Example 3: AI Chatbot for Website



Problem



MVP

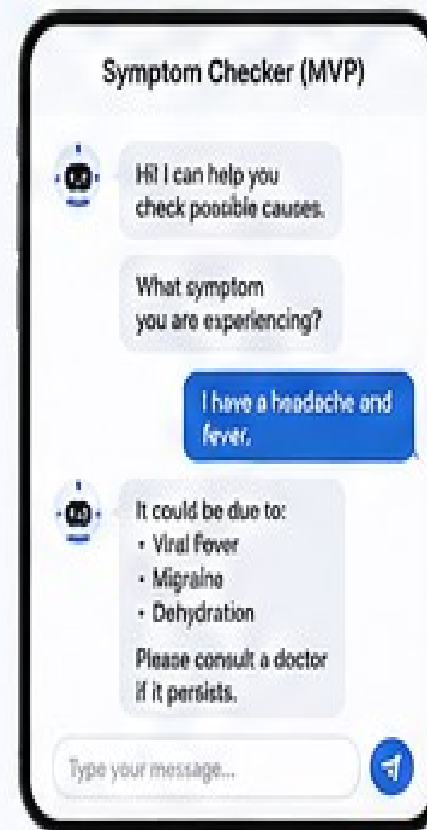


Later Full Product

4 Example 4: AI Health Symptom Checker

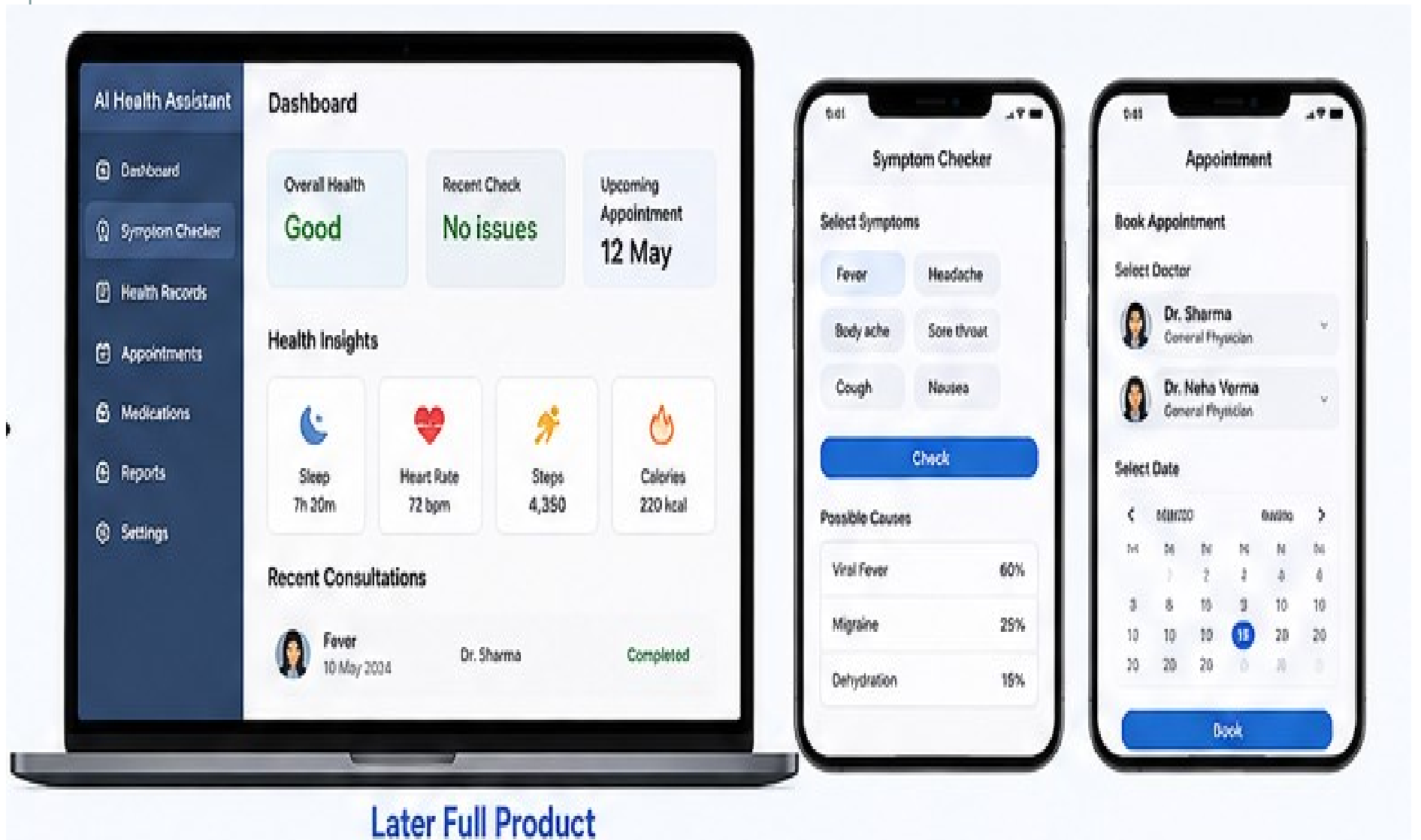


Problem

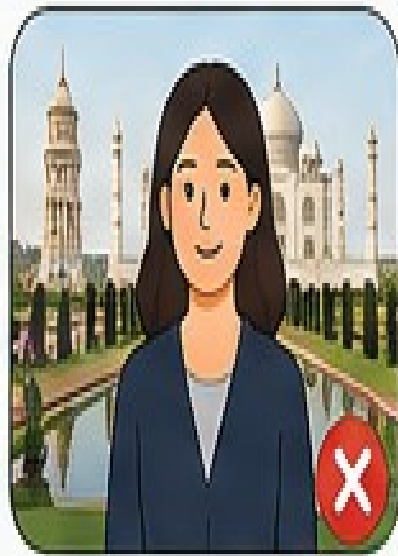


MVP

- ✓ Symptom input
- ✓ Basic AI suggestions
- ✓ General advice



5 Example 5: AI Image Background Remover



Problem

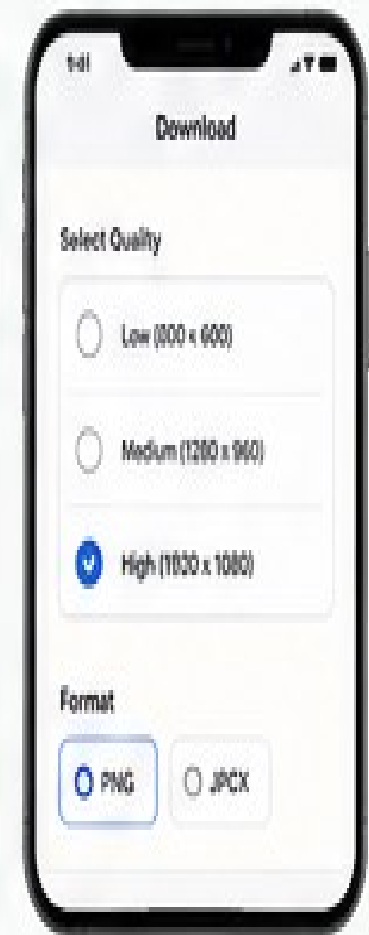
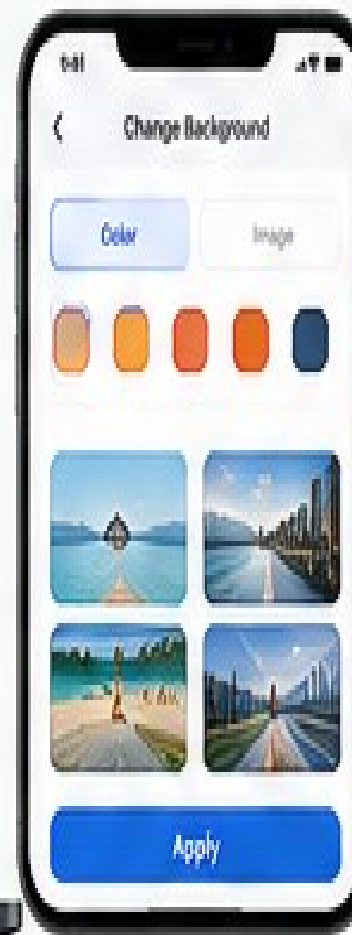
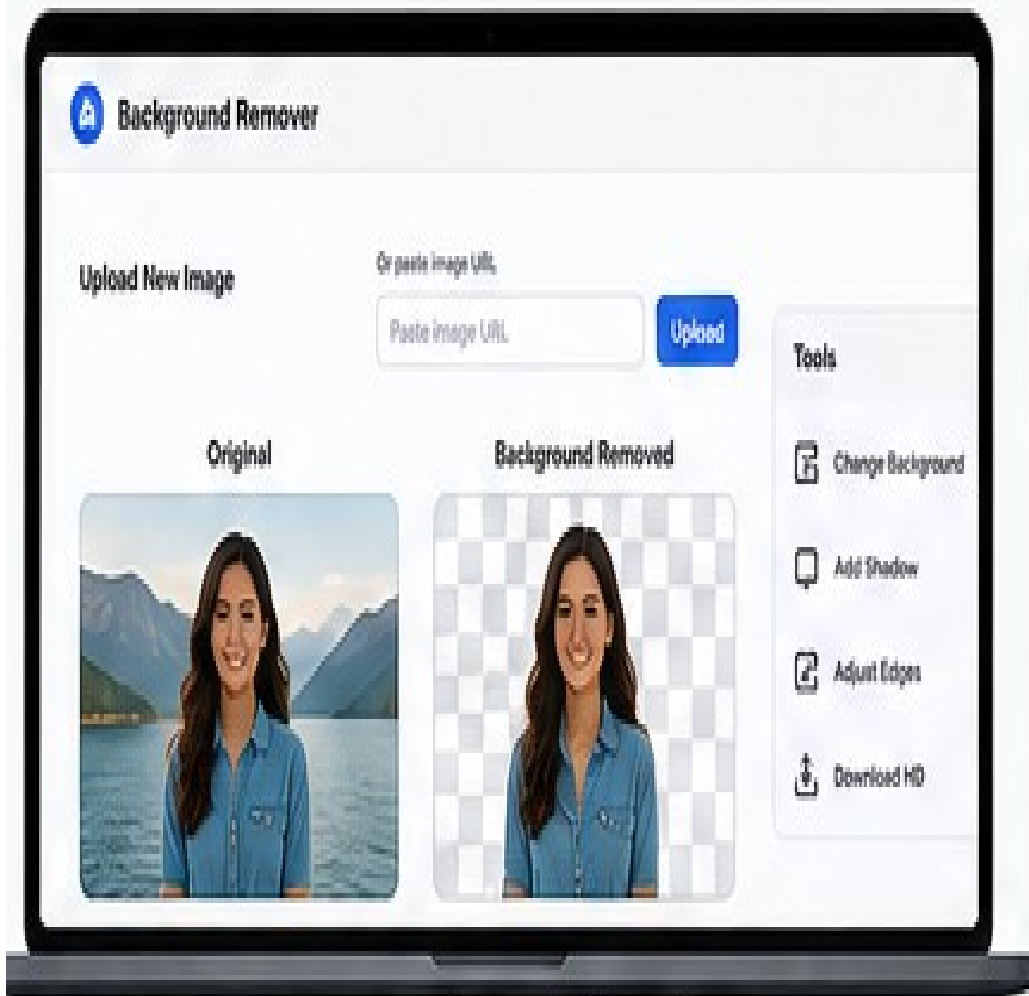


AI Background Remover (MVP)



MVP

- ✓ Image upload
- ✓ AI background removal
- ✓ Download result



Later Full Product

6 Example 6: AI Voice to Text Notes



Problem



MVP

Transcription

Meeting notes for project update. We discussed the design, timeline and next steps.

- ✓ Voice recording
- ✓ Speech to text
- ✓ Basic save

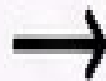


Later Full Product

7 Example 7: AI Smart Expense Tracker



Problem



MVP

Expense Tracker (MVP)

Extracted Details

Item	Coffee
Amount	₹120
Category	Food & Drinks

Save

- ✓ Receipt upload
- ✓ AI text extraction
- ✓ Save expense



Add Expense

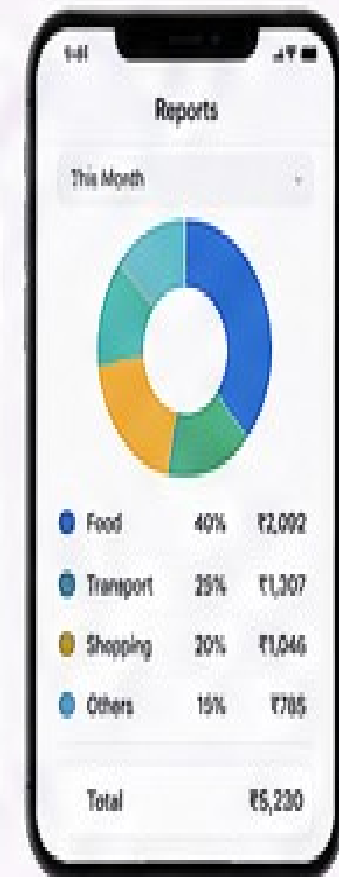
Upload Receipt

Merchant: Coffee Shop

Amount: ₹120

Category: Food & Drinks

Save



Later Full Product

Feature Prioritization



- **Feature Prioritization** is the process of deciding **which features should be developed first** in an AI product based on user needs, business goals, cost, and impact.
- It helps developers focus on the **most important features** before adding extra features.

Steps in Feature Prioritization

1

IDENTIFY USER NEEDS



Understand the target users, their problems, goals, and expectations through research, interviews, and surveys.

Output:
List of user needs and pain points

2

LIST ALL POTENTIAL FEATURES



Brainstorm and list all the possible features that can solve user problems.

Output:
Comprehensive list of features

3

EVALUATE EACH FEATURE



Evaluate each feature on key factors like user value, business impact, cost, time, and technical feasibility.

Output:
Feature evaluation scores

Steps in Feature Prioritization

4

PRIORITIZE & RANK FEATURES



Rank features from highest priority to lowest based on the evaluation results.

Output:
Prioritized list of features

5

SELECT TOP FEATURES



Choose high-priority features to include in the MVP or next release cycle based on resources and goals.

Output:
Selected features for development

6

REVIEW & RE-PRIORITIZE



Continuously review feedback, market changes, and performance metrics and re-prioritize features.

Output:
Updated priority list of features

EXAMPLE. AI STUDY ASSISTANT APP

Feature	User Value (1-5)	Business Impact (1-5)	Effort (1-5)	Total Score (Value + Impact - Effort)	Priority
 AI answers student questions	5	5	3	7	1 High
 Quiz generation	5	4	3	6	2 High
 Study reminders	4	3	2	5	3 Medium
 Progress tracking	4	3	3	4	4 Medium
 Dark mode	2	2	2	2	5 Low
 Theme customization	1	1	2	0	6 Low

Factors Affecting Feature Prioritization



- **1. User Value**

- Does the feature solve an important user problem?

- **2. Business Value**

- Will it help achieve business goals?

- **3. Development Cost**

- How much money is needed to build the feature?

Factors Affecting Feature Prioritization



- **4. Development Time**

- How long will it take to implement?

- **5. Technical Feasibility**

- Can the feature be developed with available technology and resources?

- **6. Risk**

- Does the feature introduce privacy, security, or technical risks?

Prioritization Techniques



1. MoSCoW METHOD



Must Have

Essential features for the product.



Should Have

Important but not critical.



Could Have

Nice to have, if time and resources allow.



Won't Have (Now)

Not a priority now, consider in future.

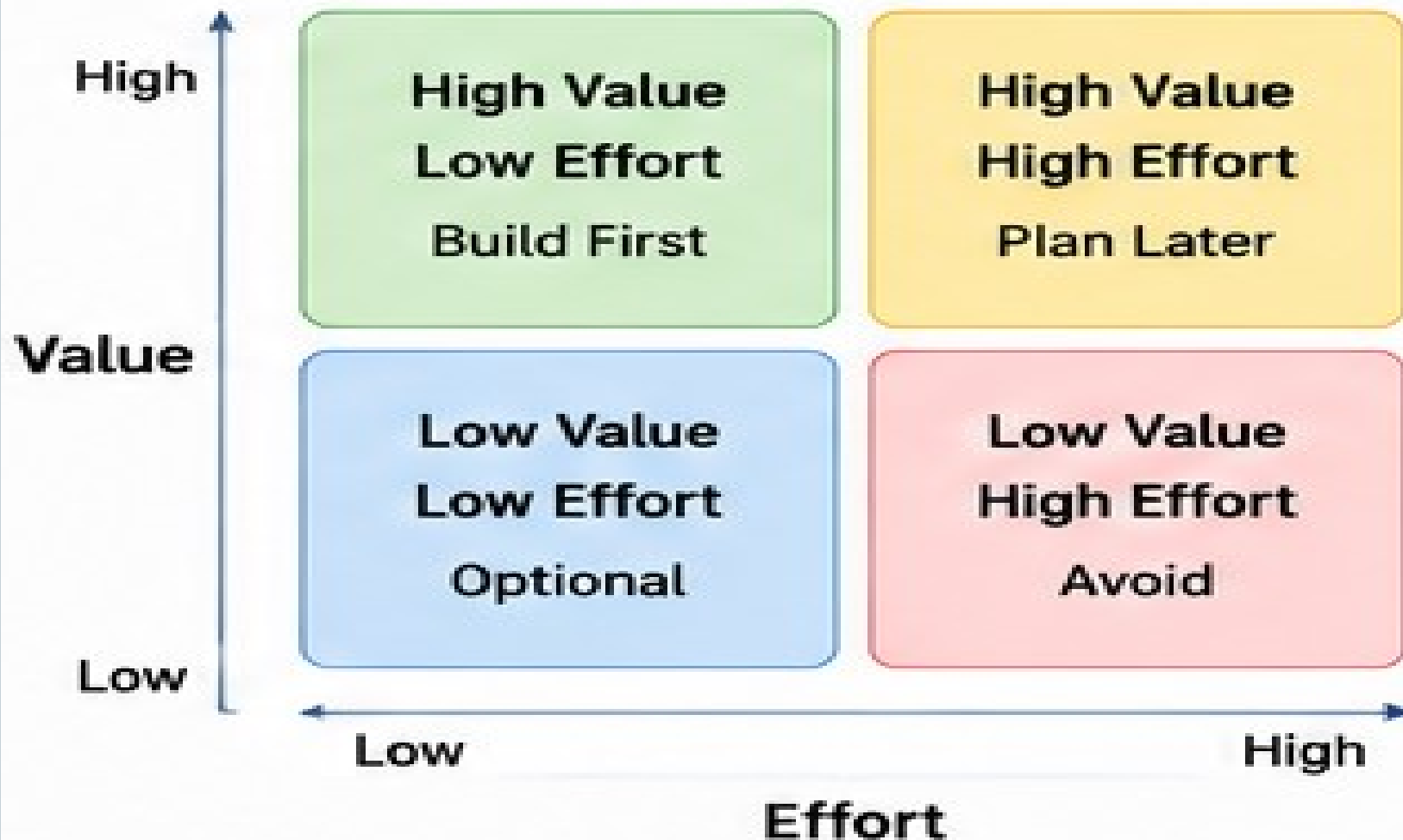
Example (AI Travel Planner)



Category	Feature
Must Have	Destination suggestions
Must Have	Budget estimation
Should Have	Hotel recommendations
Could Have	Voice assistant
Won't Have (Now)	AR travel guide

Prioritization Techniques

2. VALUE vs EFFORT MATRIX





Example Product: AI Study Assistant App

Goal: Help students learn faster, understand concepts and prepare for exams.



STEP 1: List All Possible Features

-  F1. AI Chat Tutor (Ask Doubts)
-  F2. Notes Summarization
-  F3. Practice Quiz Generation
-  F4. PDF/Book Upload & Ask
-  F5. Study Plan Generator
-  F6. Progress & Performance Analytics
-  F7. Voice to Text Notes
-  F8. Smart Flashcards
-  F9. Multi-language Support
-  F10. AI Career Guidance

STEP 2: Prioritize Using This Formula

Prioritization Score =

$$\text{User Value} + \text{Business Impact} - \text{Effort}$$

Feature	User Value (1-5)	Business Impact (1-5)	Effort (1-5)	Score (User Value + Business Impact – Effort)
F1. AI Chat Tutor (Ask Doubts)	5	5	3	7
F2. Notes Summarization	4	4	3	5
F3. Practice Quiz Generation	4	4	3	5
F4. PDF/Book Upload & Ask	4	3	4	3
F5. Study Plan Generator	4	3	4	3
F6. Progress & Performance Analytics	4	4	4	4
F7. Voice to Text Notes	3	3	3	3
F8. Smart Flashcards	3	3	2	4
F9. Multi-language Support	3	2	4	1
F10. AI Career Guidance	3	3	5	1

STEP 3: Prioritization Result



MUST HAVE (Build in MVP)

- ☒ F1. AI Chat Tutor (Ask Doubts)
- ☒ F2. Notes Summarization
- ☒ F3. Practice Quiz Generation

SHOULD HAVE (Next Phase)

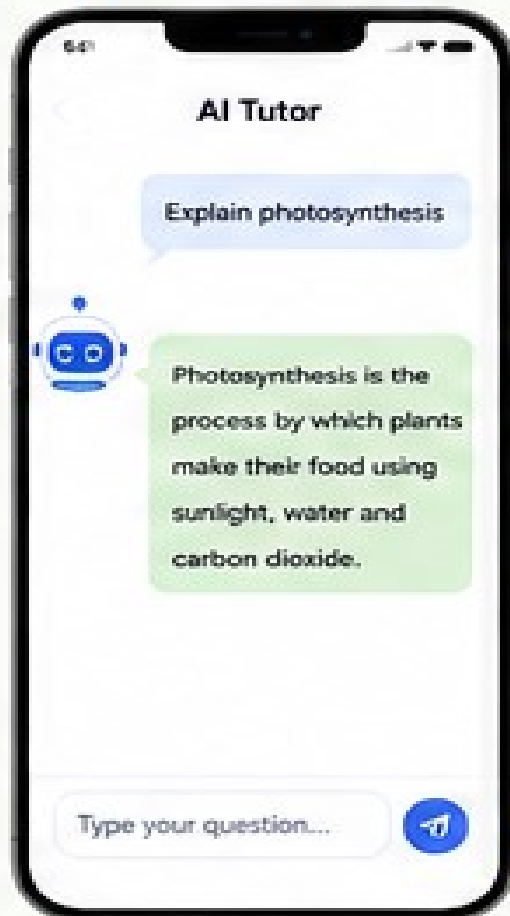
- ☐ F4. PDF/Book Upload & Ask
- ☐ F6. Progress & Performance Analytics
- ☐ F5. Study Plan Generator
- ☐ F8. Smart Flashcards

COULD HAVE (Later)

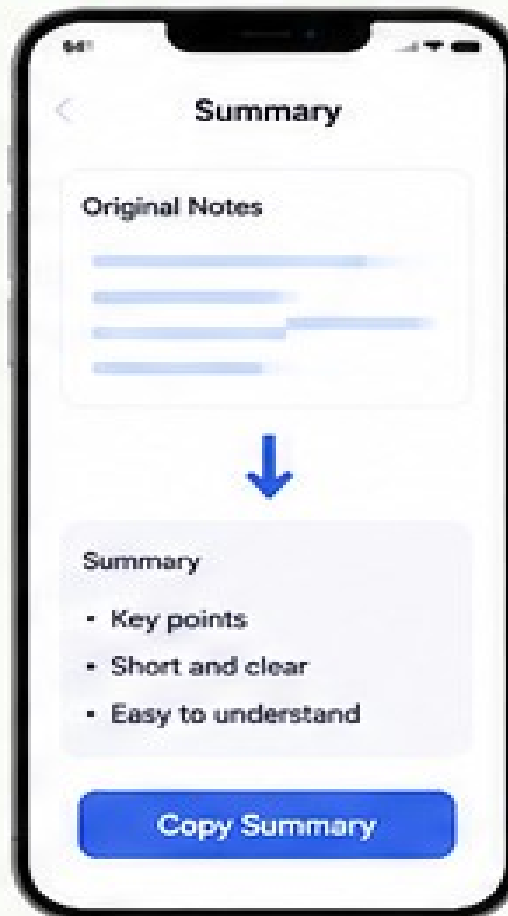
- ☐ F7. Voice to Text Notes
- ☐ F9. Multi-language Support
- ☐ F10. AI Career Guidance

MVP (First Release)

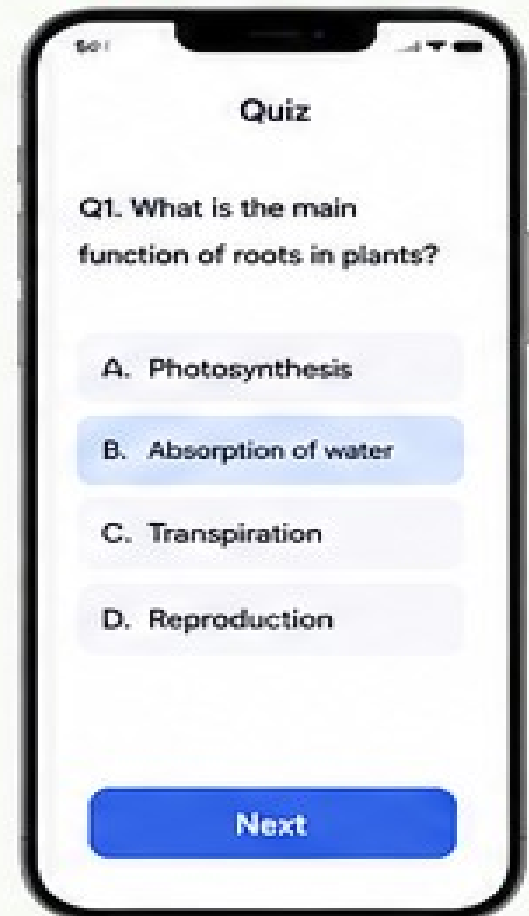
1 AI Chat Tutor



2 Notes Summarization



3 Practice Quiz



Later Full Product (After Iterations)

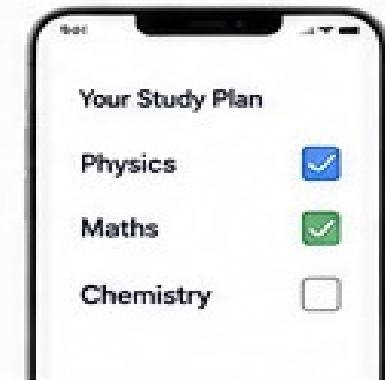
PDF/Book Upload & Ask



Progress Analytics



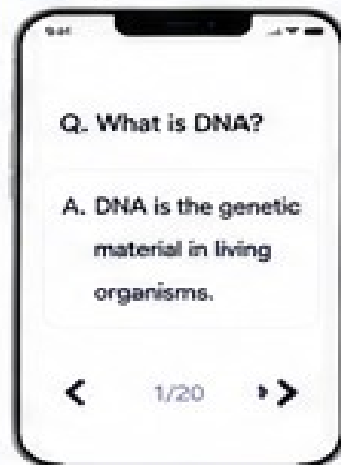
Study Plan Generator



Voice to Text Notes



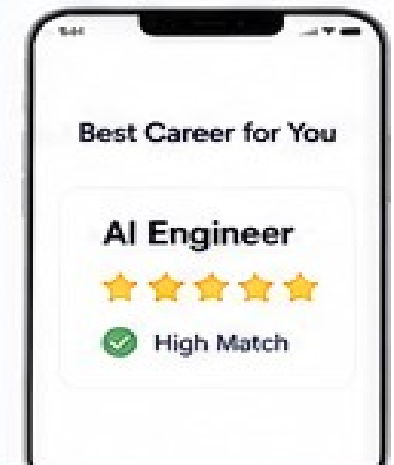
Smart Flashcards



Multi-language Support



AI Career Guidance



AI Monetization Models



- **AI Monetization** is the process of **earning money from an AI product or service** by choosing a suitable business model.

Common AI Monetization Models



- 1. SaaS (Software as a Service)
- 2. API-Based Model
- 3. Subscription Model

1. SaaS (Software as a Service)



- In the **SaaS model**, the AI software is hosted on the cloud, and users access it through the internet without installing it on their computers.
- Users usually pay a **monthly or yearly fee** to use the software.
- **Examples**
 - ChatGPT
 - Grammarly
 - Canva AI
 - Notion AI

1. SaaS (Software as a Service)



- **Advantages**

- Regular income through subscriptions.
- Easy software updates.
- Accessible from anywhere.

- **Example**

- A student pays **₹500 per month** to use an AI learning platform.

1. SAAS MODEL

(Software as a Service)



Example: Notion AI

AI writing assistant inside Notion

- Search
- Updates
- Settings
- New page

AI Writing

Ask AI

Summarize this page

Notion AI helps you write, brainstorm, edit, and summarize. It works where you work. It can draft content, fix grammar, translate, and more.

Copy

How it works



Users
(Individuals / Teams)



Notion AI
(SaaS Platform)



Cloud Infrastructure
(Hosted by Notion)

2. API-Based Model



- An **API (Application Programming Interface)** allows other applications or websites to use AI features without building their own AI system.
- Businesses pay based on **API usage** (e.g., per request or per token).
- **Examples**
 - OpenAI API
 - Google Cloud Vision API
 - Google Maps API

2. API-Based Model



- **Advantages**

- Easy integration with other applications.
- Businesses pay only for what they use.
- Suitable for developers and companies.

- **Example**

- An e-commerce website uses an AI chatbot API and pays based on the number of customer queries handled.

2. API-BASED MODEL (Pay-per-Use)



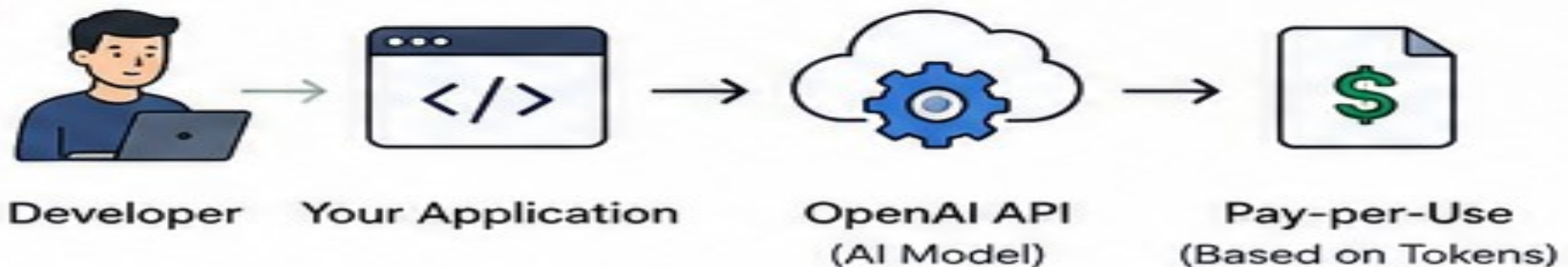
Example: OpenAI GPT API

Developers integrate GPT into their apps

```
import openai

response = openai.ChatCompletion.create(
    model="gpt-4o-mini",
    messages=[{"role": "user", "content": "Explain AI"}]
)
print(response.choices[0].message.content)
```

How it works



3. Subscription Model



- Users pay a **fixed monthly or yearly fee** to access premium AI features.
- Usually, there is:
 - **Free Plan** (basic features)
 - **Paid Plan** (advanced AI features)
- **Examples**
 - ChatGPT Plus
 - Microsoft Copilot Pro
 - Grammarly Premium

3. Subscription Model



- **Advantages**

- Predictable monthly revenue.
- Encourages long-term users.
- Easy for customers to understand.

- **Example**

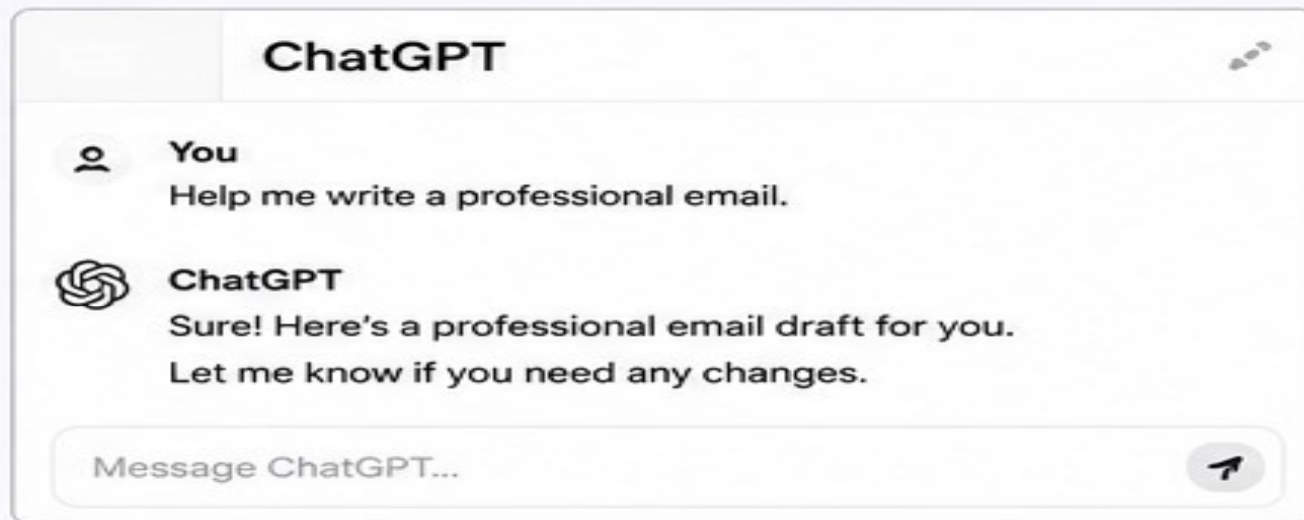
- A user pays **₹2,000 per month** to access advanced AI writing tools.

3. SUBSCRIPTION MODEL (Recurring Revenue)

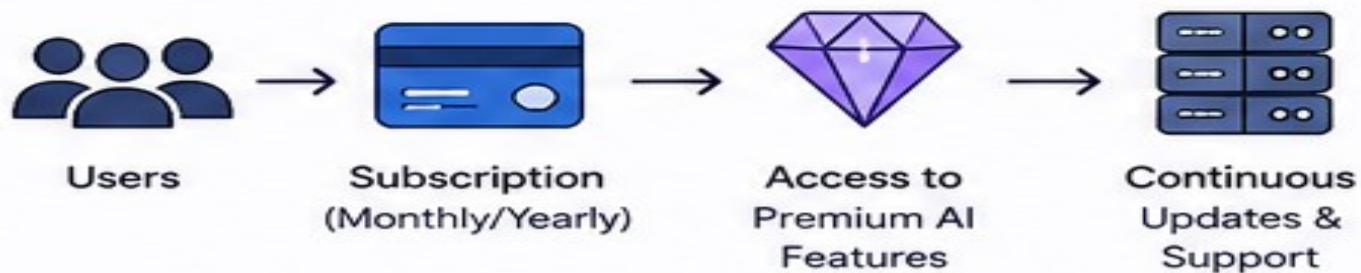


Example: ChatGPT Plus

AI assistant with premium features



How it works



Advantages of AI Monetization



- **1. Generates Revenue**
- **2. Supports Continuous Improvement**
- **3. Flexible Pricing**
- **4. Scalable Business**

Challenges of AI Monetization



- **1. High Development Cost**
- **2. Strong Competition**
- **3. User Retention**
- **4. Privacy and Security**

Real-Life Examples

AI Product	Monetization Model
ChatGPT	SaaS + Subscription + API
Grammarly	SaaS + Subscription
Canva AI	SaaS + Subscription
OpenAI API	API-Based
Google Cloud Vision AI	API-Based
Microsoft Copilot	Subscription

Business Model Canvas



- A **Business Model Canvas (BMC)** is a **one-page visual tool** used to describe how a business creates, delivers, and earns value from its products or services.
- It helps entrepreneurs and companies plan and understand their business model in a simple way.



Building Blocks of Business Model Canvas Steps

1

Customer Segments



Who are the customers?

Identify the different groups of people or organizations you want to reach and serve.

Examples

- Students
- Teachers
- Professionals

2

Value Proposition



What value do you offer?

Describe the benefits your product or service provides and the problem it solves.

Examples

- Saves time
- Improves learning
- Provides accurate results

3

Channels



How will customers reach you?

Describe the ways your product or service is delivered to customers.

Examples

- Mobile App
- Website
- Play Store / App Store

4

Revenue Streams



How will the business earn money?

Describe the sources of revenue from each customer segment.

Examples

- Subscription Fees
- SaaS (Monthly/Yearly)
- API Usage Charges
- Advertisements

5

Key Resources & Activities



What is needed to deliver value?

List the important resources you need and the key activities you perform to run the business.

Examples

Resources:

AI Models, Developers, Cloud Servers, Data

Activities:

AI Model Training, Software Development,
Testing, Customer Support

6

Cost Structure & Partners



What are the major costs and who helps?

Describe the main costs of running the business and the partners who support it.

Examples

Costs:

AI Development, Cloud Hosting, Employee Salaries, Marketing

Partners:

Cloud Provider, Data Provider, Payment Gateway, Technology Partners

Example : AI Study Assistant Business Model Canvas



Building Block	Example
1. Customer Segments	Students, Teachers
2. Value Proposition	Personalized learning, instant doubt solving
3. Channels	Mobile App, Website
4. Revenue Streams	Monthly Subscription, Premium Plan
5. Key Resources & Activities	Resources: AI Model, Developers, Cloud Servers Activities: AI Model Training, App Development, Testing
6. Cost Structure & Partners	Costs: AI Development, Cloud Hosting, Marketing Partners: Cloud Provider, Payment Gateway, Educational Content Providers

Introduction to API Concept



- **API (Application Programming Interface)** is a set of rules that allows **two software applications to communicate and exchange data** with each other.

How API Works



(1) You ask ChatGPT: "Translate 'Hello' to French.“

(2) App sends request through API.

(3) AI processes the request.

(4) AI returns: **Bonjour.**

Everyday Examples of APIs



- Google Maps API → Shows maps in apps.
- Weather API → Displays weather information.
- Payment API → Online payments.
- AI API → Chatbots, translators, image generators.

Advantages of API



- Easy communication between applications.
- Saves development time.
- Reuses existing AI services.
- Easy integration.

Understanding AI APIs (OpenAI API – Concepts Only)



- An **AI API** allows developers to use AI capabilities such as text generation, translation, summarization, or image generation in their own applications.
- Instead of building an AI model from scratch, developers connect to an existing AI service.

How AI API Works



- (1) Your app captures user input (e.g., "Translate 'Hello' to Spanish").
- (2) Your app wraps this text into an API call and sends it to OpenAI's servers.
- (3) OpenAI's neural network processes the request.
- (4) OpenAI sends back the response (e.g., "Hola").
- (5) Your app displays "Hola" to the user.

Common Uses of AI APIs



- AI Chatbots
- Language Translation
- Text Summarization
- Image Generation
- Speech Recognition
- Question Answering

Basic API Key and Security Awareness



- API Key : An **API Key** is a **unique secret code** that identifies and authorizes an application to use an API.
- It works like a **password** for accessing AI services.

Security Awareness for API key



- **Never Hardcode:** Do not paste your API key directly inside your source code files.
- **Use Environment Variables:** Store keys securely in environment files (like .env) that stay on your local machine.
- **Never Commit to GitHub:** Ensure your .env files are added to your .gitignore file so they are never leaked online.
- **Rotate Keys:** Change and regenerate your API keys periodically to minimize exposure risks.



Real-Time Example for API Key security

Example 1: Weather App



- A weather app uses a **Weather API**.
- **What Happens?**
Weather App → (API Key) -> Weather API
Someone steals the API key
- **Result**
They can make unlimited weather requests.
The owner may exceed the free usage limit.
Extra charges may apply.
- **Lesson**
Keep the API key secret.

Example 2: Student Project



- A student builds an AI chatbot project and uploads the code to GitHub.

What Happens?

Student Project → GitHub (Public) -> API Key Visible

A stranger copies the API key.

Result

Uses the AI service for free.

Student's API credits are exhausted.

The project stops working.

Lesson: Never upload API keys to GitHub or public repositories.



Cost Awareness in AI Products (Token Concept)



- A **token** is a small unit of text processed by an AI model.
- AI APIs usually charge based on the number of **tokens** used rather than the number of questions.

Example



- Sentence:

"AI helps students learn."

- Approximate tokens:

AI | helps | students | learn | .

≈ 5 tokens

- More tokens = More processing = Higher cost

Short Prompt



Example 1: Translate "Hello" to French.

Input (Prompt):

Translate "Hello" to French.

Approx. Input Tokens:

Translate "Hello" to French .

≈ 5 tokens

Output (Response):

Bonjour

Approx. Output Tokens:

Bonjour

≈ 1–2 tokens



Low Cost

Daily Usage Example

Users	Avg. Tokens per conversation (Input + Output)	Conversations per day	Total Tokens per day	Estimated Daily Cost (GPT-3.5 Turbo)
100	150	200	$100 \times 200 \times 150$ $= 3,000,000$	$3,000,000 / 1000 \times \0.002 $= \$6.00$ per day

More tokens



More cost



Ways to Reduce AI Cost



- Use short prompts.
- Request concise answers.
- Avoid unnecessary data.
- Cache repeated responses.



Basic AI Evaluation Methods



- AI evaluation checks whether an AI system produces useful, accurate, and reliable results.
- Output quality checking
- User feedback
- Simple testing checklist

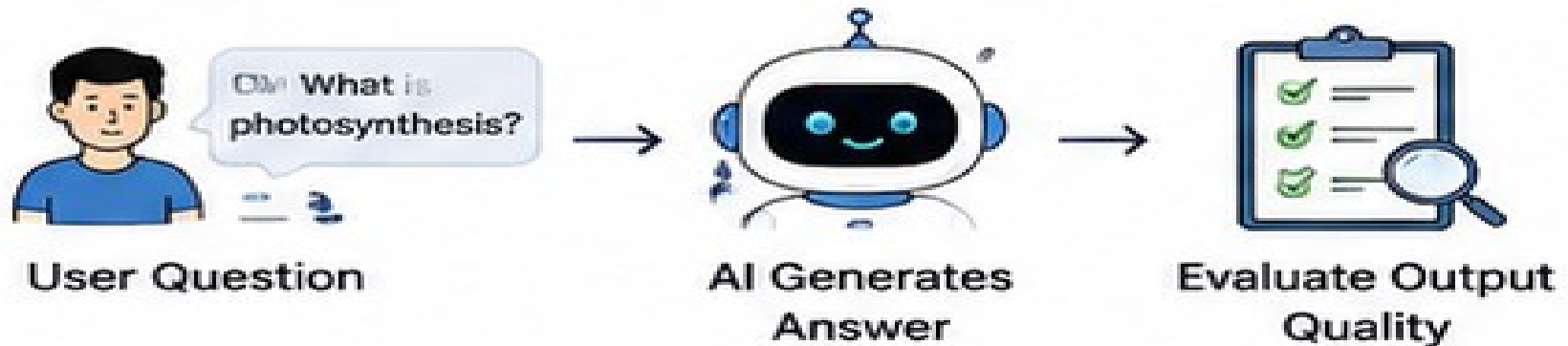
(1) Output quality checking



- Checking whether AI responses are correct, relevant, complete, and understandable.
- **Questions to Ask**
- Is the answer correct?
- Is it relevant?
- Is it easy to understand?
- Is it complete?

Example

How it works



Example

Good Output ✓

Photosynthesis is the process by which green plants use sunlight, water and carbon dioxide to make their own food (glucose) and release oxygen.

Poor Output ✗

Photosynthesis is when plants eat food and grow. It happens at night and makes plants bigger.

(2) User Feedback



- Collecting opinions from users to improve the AI system.
- **Methods**
- Ratings ☆ ☆ ☆ ☆ ☆
- Thumbs Up / Down  
- Feedback Forms
- Comments
- Surveys

Example



How it works



User Uses
AI System



How was your
experience?



Feedback form

Submit

User Gives
Feedback



Feedback is
Analyzed

Example

AI Chatbot App

Rate your experience



What did you like?

Quick answers, helpful



What can be improved?

Sometimes answers are too long.

Feedback Types

- ✓ Star Ratings (1 to 5)
- ✓ Thumbs Up / Down
- ✓ Comments
- ✓ Feedback Forms
- ✓ Surveys

(3) Simple Testing Checklist



- Before releasing an AI product, developers perform a simple checklist.

How it works



Prepare
Checklist

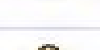


Test AI System
Using Checklist



AI System is
Ready to Use

Example Checklist

Checklist Item		Yes / No
 Does AI give correct answers?		
 Is the response relevant to the question?		
 Is the output easy to understand?		
 Is the system fast and reliable?		
 Is user data secure and private?		
 Does AI avoid biased or harmful content?		
 Does the application work properly on all devices?		
 Are error messages user-friendly?		

3-4 Marks Questions



- Explain the **Data Collection and Deployment** stage with an example.
- What is meant by **Retirement or Replacement** of an AI product?
- Why is **MVP important** in AI Product Design?
- Differentiate between **MVP and Full Product**.
- Explain MVP with a **real-time example**.
- Why is **Feature Prioritization** important in AI Product Design?
- Short Note on : MoSCoW Method for feature prioritization.
- Compare the **SaaS, API-Based, and Subscription Models** in AI Product Design.
- Give one example each of **SaaS, API-Based, and Subscription** models.
- Explain the role of **Channels** in the Business Model Canvas.
- Define **API (Application Programming Interface)**. Explain how an **API works** with a simple diagram.
- Explain AI APIs using an **AI chatbot** example.
- Why is an API Key important? List any **four best practices** for API Key Security.
- Explain the **token concept** with a simple example. Give any **four ways to reduce AI API costs**.
- Explain AI evaluation using an (1) **AI chatbot** (2) **AI Medical Assistant** example.
- Differentiate between **Output Quality Checking** and **User Feedback**.

7 marks Questions



- Explain the **stages of AI Product Lifecycle**.
- Describe the complete lifecycle of an **AI Study Assistant** from idea generation to retirement.
- Define **Minimum Viable Product (MVP)** and explain its importance in AI Product Design with a neat diagram.
- Describe how an **AI Medical Assistant** is developed using the MVP approach.
- What factors should be considered while prioritizing AI product features?
- Define Feature Prioritization and explain its techniques in detail.
- Explain the **steps involved in Feature Prioritization** with a suitable real-time example.
- Explain the **User Value + Business Impact – Effort** scoring method with a suitable example.
- Define **AI Monetization** and explain the different **AI Monetization Models** with suitable examples.
- Explain the **building blocks of the Business Model Canvas** with suitable examples.
- Draw and explain the **Business Model Canvas** for an **(1) AI Study Assistant (2) AI travel planner**.
- Explain **API Key Security Awareness**. What problems can occur if an API Key is exposed?
- Define **AI Evaluation** and explain the **basic AI evaluation methods** with suitable examples.